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(57) Abstract:

The present invention describes the design and analysis of a compact dual-band symmetric 2×1 circular CPW-fed array antenna for advanced wireless communication applications. The antenna is realized on a flexible polyimide substrate (ϵ_r 3.5, thickness 0.1 mm), making it suitable for compact and conformal devices. A coplanar waveguide (CPW) feeding technique is employed to achieve improved impedance matching, reduced radiation loss, and simplified fabrication. The proposed antenna operates at a primary resonant frequency of 2.66 GHz, achieving return loss of -36.46 dB and a VSWR of 1.03, ensuring efficient power transfer and minimal signal reflection. Additionally, the antenna exhibits a secondary resonance at 7.93 GHz with a return loss of -18.91 dB and a VSWR of 1.25, enabling reliable dual-band performance. The design achieves a wide impedance bandwidth of 480 MHz (2.38–2.86 GHz) for WLAN applications and 210 MHz (7.82–8.03 GHz) for satellite applications. The incorporation of slot structures and array configuration enhances bandwidth, gain, and multiband behaviour. The antenna provides a maximum gain of 8.13 dB at 2.66 GHz and 12.89 dB at 7.93 GHz with stable radiation characteristics across both bands. Due to its compact size, lightweight structure, dual-band capability, wide bandwidth, and high efficiency, the proposed antenna is highly suitable for WLAN systems, satellite communication, and modern wireless communication devices, including portable and embedded systems.